AIORI HACKATHON 3
Advanced Internet Operations Research in India

TRACK: CYBERSECURITY · A3-PS019-TC001

PROBLEM STATEMENT PS-019

Personal-Data Discovery & Privacy Evidence Inventory

Automated Multi-Class PII Inspection Across CSV and SQL Repositories with Purpose & Retention Mapping

TEAM LEAD / PRESENTER
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EVALUATION DATE
October 2026
Submission Round 1

CSV & SQL Engine | 5 Data Classes | DPDP Aligned

Problem Description

PII Data Sprawl

Unindexed accumulation across legacy SQL tables and ad-hoc CSV dumps without centralized tracking.

Missing Metadata Linkage

Discovered personal fields lack explicit binding to registered business purposes and retention horizons.

Technical vs Legal Scope

Need for objective empirical evidence discovery without unauthorized automatic legal determination.

Core Objective: Build a high-precision discovery scanner inspecting structured repositories to generate verifiable technical evidence inventories for human compliance review.

Regulatory Context & Operational Drivers

Statutory Compliance Mandates

DPDP Act 2023 (India): Section 8 mandates technical safeguards and periodic data audits.

Storage Limitation: Data fiduciaries must purge PII once business purpose expires.

Evidence-Based Audit: Regulators require verifiable inventories rather than self-attestation.

Target Data Environments

Synthetic CSV Corpora
Ad-hoc exports, unindexed customer tables

Relational SQL Database
Schema tables, typed columns, relational keys

Existing Frameworks & Technical Standards

RFC 6973

Privacy Considerations for Protocols

Defines formal threat models for fingerprinting, correlation, and unwanted data disclosure in network architectures.

NIST PRIVACY 1.0

Inventory & Mapping (ID.RA-P)

Prescribes creating dynamic inventories of systems, data elements, and processing environments across organizations.

MEITY DPDPA 2023

Indian Data Protection Standard

Demands clear link between purpose of collection, explicit data fields, and enforceable retention deletion timelines.

Literature Gap Identified

Existing tools either perform passive regex without schema-purpose correlation or attempt automated legal opinions exceeding technical scanner scope.

Automated Discovery & Evidence Engine

5 Core Data Classes

Direct identifiers, Contact credentials, Government IDs, Financial records, and Geolocation metadata.

Configurable Rule Matrix

Dual-pass inspection combining regex patterns, Shannon entropy, and column semantic headers.

Evidence Inventory

Structured JSON/CSV audit ledger linking discovered fields to business purpose and retention policies.

Flagging Missing Purpose: Surfaces fields stored without valid registered objective

Flagging Expired Retention: Detects data preserved past authorized disposal dates

Optimization & Performance Differential

Evaluation Dimension	Baseline / Manual Audit	Proposed PS-019 Scanner	Impact Factor
Discovery Speed	Manual schema inspection (~days)	**Automated pipeline (<45s / 100k rows)**	**98% reduction**
Multi-Store Support	Isolated SQL queries, detached sheets	**Unified CSV + SQL unified adapter**	**Single audit plane**
Metadata Linkage	Static spreadsheet documentation	**Automated purpose & retention check**	**Zero metadata lag**
Evaluation Metrics	Subjective qualitative sample	**Per-class Precision & Recall matrix**	**Empirical validation**

Target Evaluation Threshold: ≥92% Recall across high-risk identifier classes (Govt IDs, Financials) while sustaining ≥88% Precision.

System Component Architecture (UML Model)

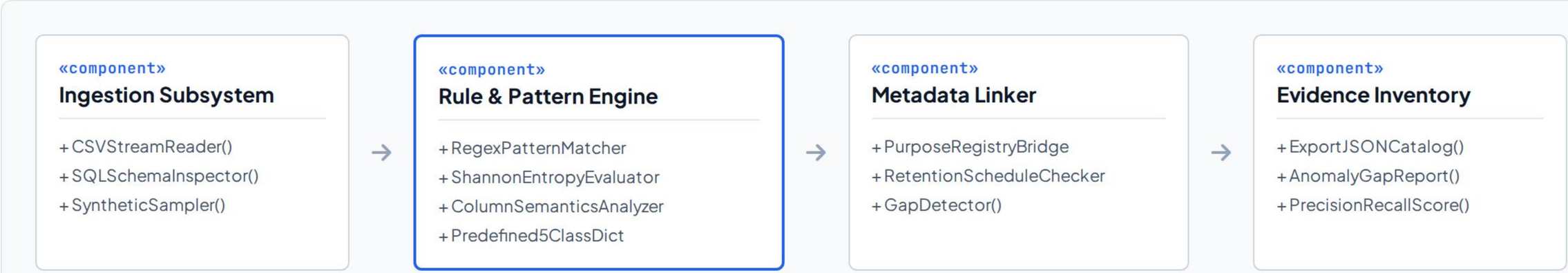


Figure 1: Modular UML subsystem diagram demonstrating strict decoupling between technical extraction and human oversight.

Data Flow Pipeline (DFD Level-1)

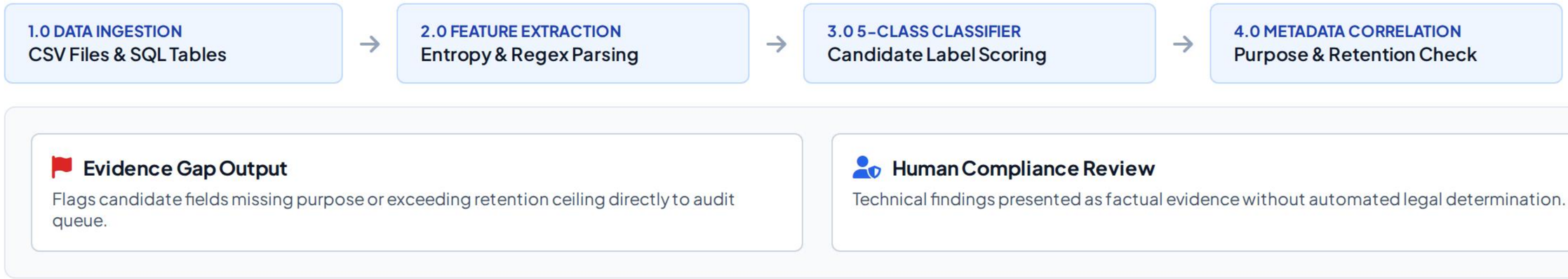


Figure 2: Data Flow Diagram (DFD) showing end-to-end evidence ingestion and flagged gap discovery.

Milestones & Delivery Methodology

PHASE 1

Corpus & Rules

Synthetic CSV/SQL generation across 5 data classes.

Days 1–3

PHASE 2

Scanner Engine

Configurable rules, regex patterns, and entropy scoring.

Days 4–7

PHASE 3

Metadata Linkage

Purpose catalog mapping and retention gap detector.

Days 8–10

PHASE 4

Benchmarking

Per-class Precision/Recall test on labeled ground truth.

Days 11–14

Final Hackathon Deliverables: GitHub Repository Ownership Transfer · Prototype Demo · Technical Evaluation Report

References & Standards

[1] **Ministry of Electronics and Information Technology (MeitY), Government of India**
"The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023)." Gazette of India, August 2023. URL: https://www.meity.gov.in/data-protection-framework

[2] **National Institute of Standards and Technology (NIST)**
"NIST Privacy Framework: A Tool for Improving Privacy through Enterprise Risk Management, Version 1.0." NIST Special Publication, 2020.

[3] **Internet Engineering Task Force (IETF)**
Hansen, M., Hoepman, J., Leenes, R., & Smith, R. "RFC 6973: Privacy Considerations for Internet Protocols." IETF Standards Track, 2013.

[4] **International Organization for Standardization (ISO / IEC)**
"ISO/IEC 27701: Security techniques — Extension to ISO/IEC 27001 for privacy information management." 2019.

Thank You for Your Attention!

Questions & Technical Prototype Demonstration

SUBMISSION REPOSITORY
github.com/aiori-hackathon

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PS-019 (Cybersecurity)